

Sustainability scenarios

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What is sustainability?

A societal definition:

"Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs."

"Brundtland Definition"

from WCED 1987: Our common future

A corporate definition:

"A business approach to create long-term shareholder value by embracing opportunities and managing risks deriving from economic, environmental and social developments." -- *Dow Jones Sustainability Group Index*

“In a **stable economy**, sustainability is the competitive advantage strategy.

In a **down economy**, sustainability is the turnaround strategy.

In a **collapse**, sustainability is a survival strategy.”

Hunter Lovins, 2009

What are Scenarios?

- Scenarios are possible views of the world, described in narrative form (stories) that provide a context in which managers can make decisions.
- By seeing a range of possible worlds, decisions will be better informed, and a strategy based on this knowledge and insight will be more likely to succeed.
- Scenarios do not predict the future, but they do illuminate the drivers of change: understanding them can only help managers to take greater control of their situation.



The 2050 predictions...



→ Our lives are undergoing dramatic changes

Population growth



New technology



Unlimited choice



Information explosion



Freedom to connect



Resource depletion





Natural resources

are being depleted at record rates

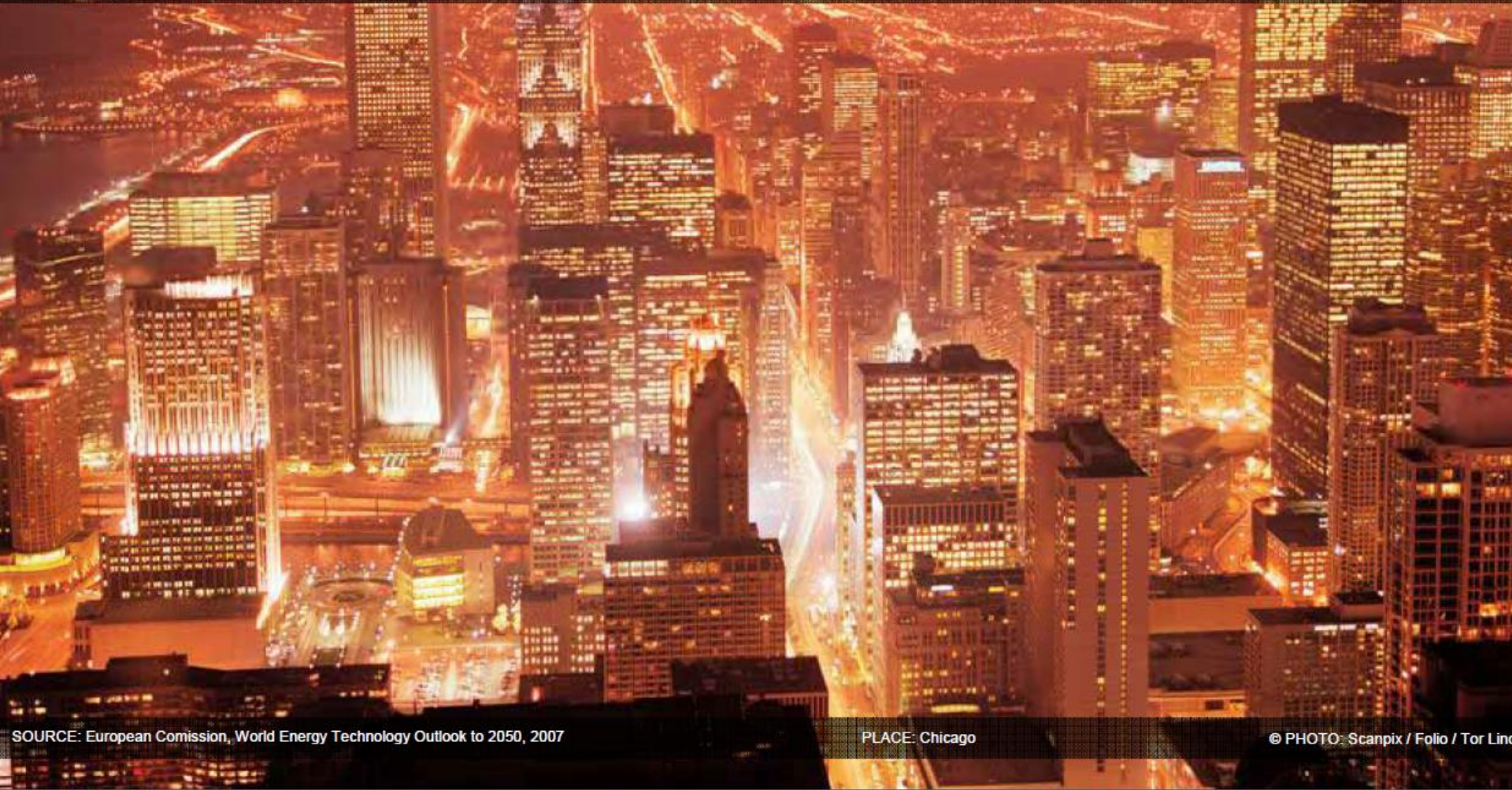
The image features two partial views of the Earth from space, one on the left and one on the right, showing blue oceans, white clouds, and green landmasses. The background is black. In the center, the text '2xEarths'' is written in a large, white, sans-serif font.

2xEarths'

By 2050 we will be using
TWICE as many natural
resources as the Earth
can replenish.

200%

World demand for energy is expected to more than DOUBLE.





300%

The number of cars in the world is expected to TRIPLE.

400%

The size of the world economy
(GDP) is expected to
QUADRUPLE.

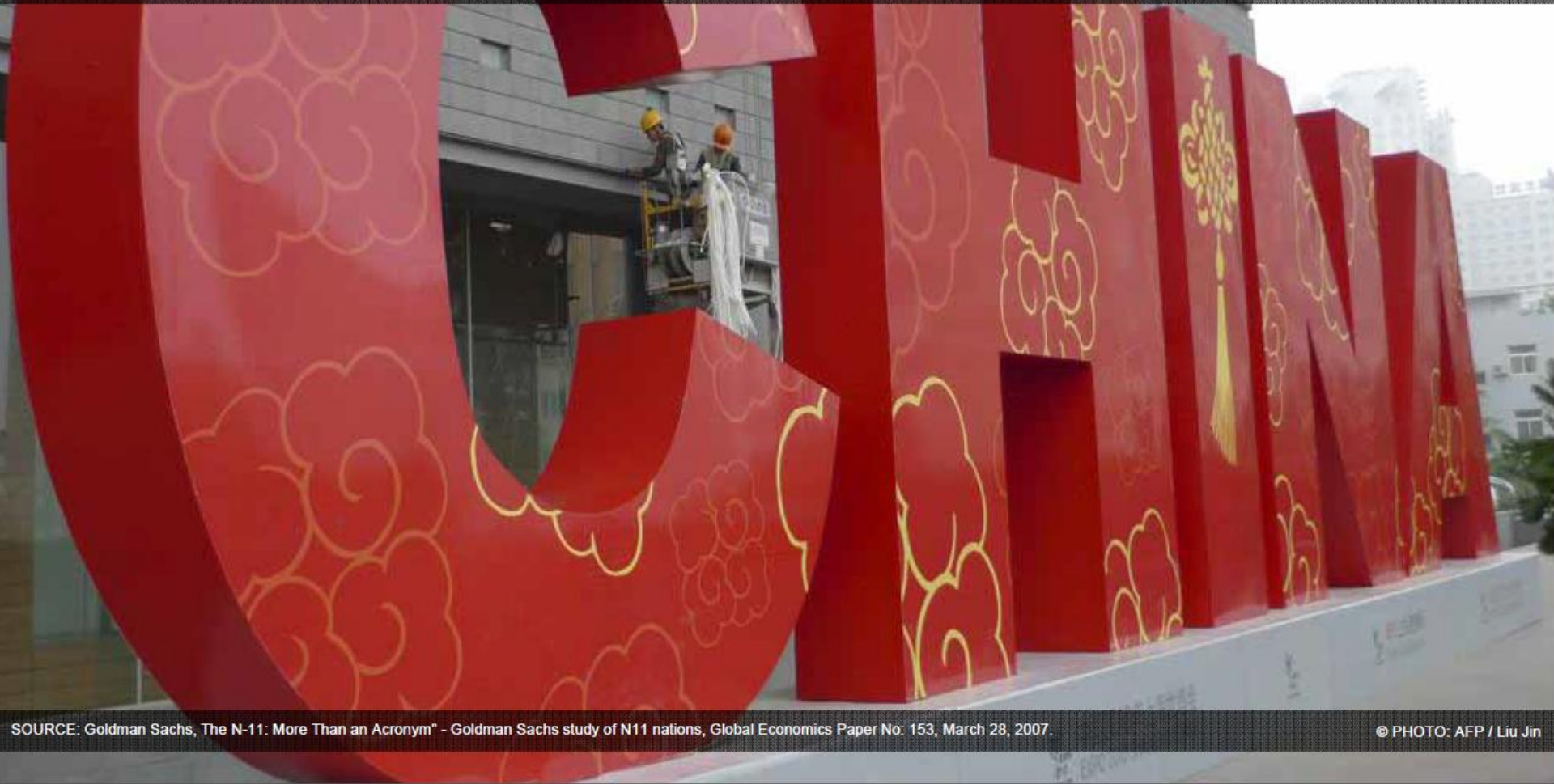


Parts of the world will grow and PROSPER.

1,5 billion

Projected population in China.

China is expected to become the world's **LARGEST** economy.





1,6 billion

Projected population in India.

India is expected to become the **THIRD** largest economy in the world.

Other parts of the world will be DEVASTATED.



Crude

Over the past 50 years,
we have changed ecosystems more extensively
than in any period in human history.

90%

of the large fish in the seas have been plundered.



75%

of the natural ecosystems on which we depend
have already been degraded.





50%

of all the forests that once covered the Earth,
have been cleared for farmland and urban development.

A dramatic underwater photograph of a seabird, possibly a booby, entangled in a fishing net. The bird is suspended in the water, its wings and legs caught in the mesh. Sunlight filters down from the surface, creating a shimmering effect on the net and the water. The overall tone is somber and urgent, highlighting the impact of overfishing on marine wildlife.

25%

of all species is heading
for extinction.

The competition for FOOD, WATER, LAND and ENERGY will intensify.



EVENT: Food Crisis in Kolkata, West Bengal, India

© PHOTO: Scanpix / ZUMA press



2 billion
people are already affected by water scarcity.

A large crowd of people is crossing a bridge over a river. The bridge is made of wooden planks and is supported by large, dark, rounded pillars. The water is calm, and the sky is overcast. The crowd is dense, with many people standing on the bridge and others waiting on the riverbank. The scene is captured in a wide shot, showing the scale of the migration.

Migration

Damage from climate change forces people to leave hard-hit rural areas to seek better conditions in already over-crowded cities.

An aerial photograph showing a vast, dense slum in Lagos, Nigeria. The foreground and middle ground are filled with a tightly packed cluster of small, low-rise buildings with flat roofs, many of which appear to be made of corrugated metal or similar materials. The buildings are closely situated, with narrow, unpaved paths visible between them. In the background, a tall, modern skyscraper with a distinctive blue and white facade stands out against the hazy sky. The contrast between the informal settlement and the modern high-rise is a central theme of the image.

1 billion

are currently living as slum dwellers
in the developing world's cities.

A photograph showing a massive crowd of people wading through deep floodwaters. Many individuals are holding open umbrellas, suggesting it is raining. Several trucks are partially submerged in the water, with people standing on their roofs or in the back of their cargo areas. The scene depicts a significant displacement of people due to flooding.

200 million

Estimated number of climate refugees by 2050.

The world consumption of oil...

World's proven oil reserves 1200 billion barrels





+

600 billion barrels
to extract from tar sands
in Canada and Venezuela

With a world oil consumption
growth of 1% per year ...

...the oil reserves will be exhausted in
less than

40 years



**OUR ENERGY CONSUMPTION:
107,000 TRILLIONS KWH THIS YEAR**



**ELECTRICITY PRICE RISES:
NEARLY MORE THAN 30% EVERY YEAR**



**DESERTS GROWING UP:
30,000 HECTARES A DAY MORE DRY LAND**

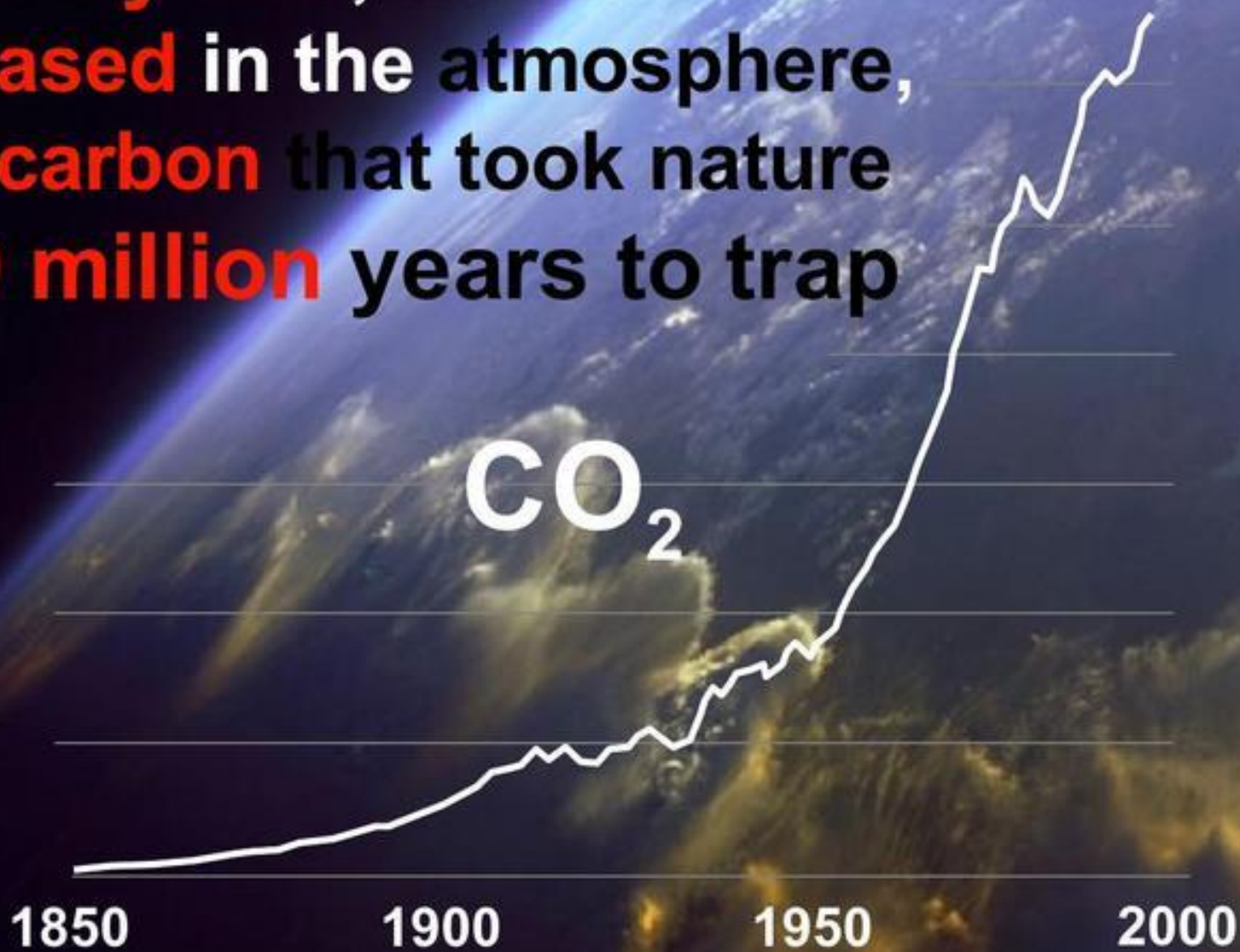


**WATER IS GETTING LESS:
2.5 BILLIONS PEOPLE WILL DIE OF THIRST**



**The sea level has risen 15 to 20 cm
over the 20th century and is going to rise
another 18 to 59 cm by the year 2100**

In **200 years**, we will have
released in the atmosphere,
the **carbon** that took nature
600 million years to trap





Unsustainable

The fashion and speed in which we keep exploiting Earth,
is simply unsustainable in the long term

Many people are aware of the state of the world and try to live responsibly...

...but only **2%** buy green.

They buy from considering:

1. Price
2. Quality
3. Ease of use
4. Availability
5. **Eco friendliness**

Our challenge is to
make it affordable,
worth while, inspiring,
rewarding and fun, to
live sustainably.

‘Design is not just about making something, it is about **designing the feelings** of the person who uses it,’ says **Tokuji Yoshioka**, a Japanese designer

“In our scenarios we need to look for a positive direction for the consumer – saving money, doing good, getting a better life, and having fun! We want people to seek out sustainable solutions and have long lasting enjoyment from them...”

Scenario 1

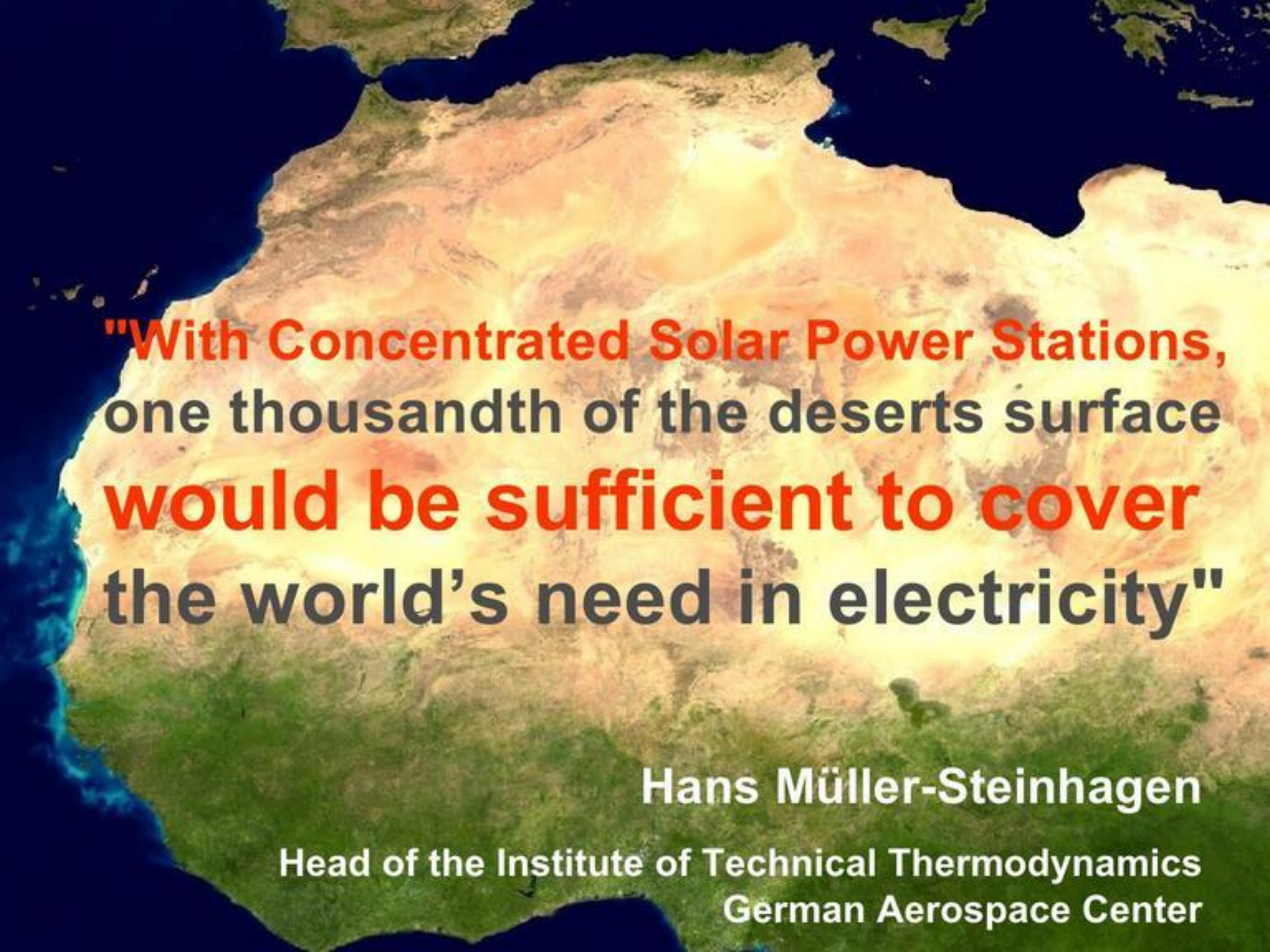
Solar energy



**THE SUN COMPENSATES
1,000 TRILLION NUCLEAR REACTORS**



In only one day, the sun provides the Earth with **the same energy** to that consumed by humanity **in 35 years**

A satellite map of the African continent, showing the vast desert regions in shades of orange and yellow, and the surrounding green landmasses and blue oceans. The map is centered on the continent, with the Atlantic Ocean to the west and the Indian Ocean to the east.

**"With Concentrated Solar Power Stations,
one thousandth of the deserts surface
would be sufficient to cover
the world's need in electricity"**

Hans Müller-Steinhagen

**Head of the Institute of Technical Thermodynamics
German Aerospace Center**



**MAKE YOUR OWN POWER:
GREEN, CLEAN AND CHEAP**

Mobile solar-powered hospital (with refrigerator)



In this future scenario Solar energy is provided by enormous power plants (like the one today in the Libyan desert) that can provide enough energy for the whole of Europe.

Our opportunity is to provide small scale solar units for individuals, built into homes. Encompassing energy provision, refrigeration, cooking, solar blinds, roofs and facade panels. At a price which is affordable to the many people.



Scenario 2

ECO x ECO

Traditional thinking says

eco = expensive.

Ecologically friendly

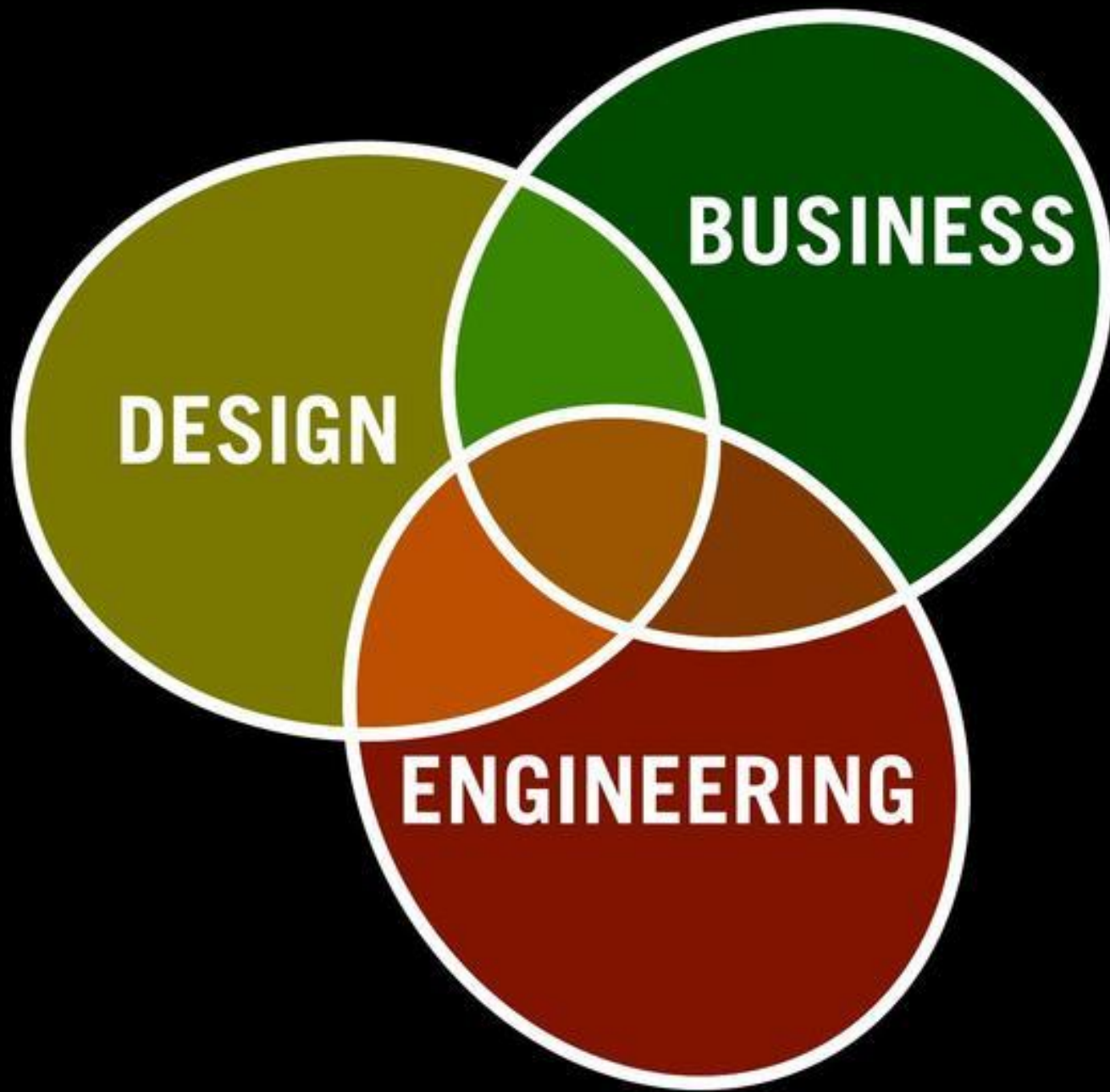
ECO x **ECO**

Economically viable

maximizing eco-efficiency
= minimizing expenses

...with one big if:

**it must be designed from
the start.**



Two basic strategies:

1) Design for longevity

2) Design for disposal

(if the product is necessary **at all)**

Design for longevity

- Timeless, classical design
- Long lifespan
- Durability
- Designed for ease of repair and maintenance
- Designed for modularity
- Possibilities for reuse
- Possibility to upgrade with latest technology

Design for disposal

- Fashionable design
- Designed for takeback
- Designed for recycling
- Designed for eco-friendly disposal

By 2050,

93% reduction

namely,

everything needs to be
redesigned.

75% reduction means

2 x endurance

2 x less inputs

If you have:

a) new technologies

b) new user needs

c) new markets

d) new business plans...?

...what you get is

**new opportunities to
get cleanly rich**

(as opposed to filthy rich)

Other hot ideas:

Breaking the cost barrier

Constraint-based design

Cradle to Cradle

This scenario is about designing our way out of the current situation. Re-designing most things over time, finding new materials, process, designs that lead to second designs and seeing waste as a value chain.

It's also about collaborative consumption – sharing, renting, bartering, trading.

It's about open source collaboration and the "long-tail" model of retailing.

It means a new design approach, new services, new materials and new production models...



IKEA PS ELLAN



Scenario 3

Eco points

2025 future scenario: Involving and rewarding green behaviour

It's the year 2025. As LCA and value-chain assessments have grown more robust and detailed over the years, so too has the level of transparency and accountability in the marketplace.

The methods by which we measure and track environmental data have become so effective that every individual around the world can measure his or her environmental footprint.

Worldwide, an Eco-Point system has been implemented to help people track, WeightWatchers style, their impact on the environment so that they can budget their consumption habits and behaviors accordingly.

Flick on a light-switch? 2 Points. Head into the office for work? 12 Points. Read the news online instead of in print? -1 Point.

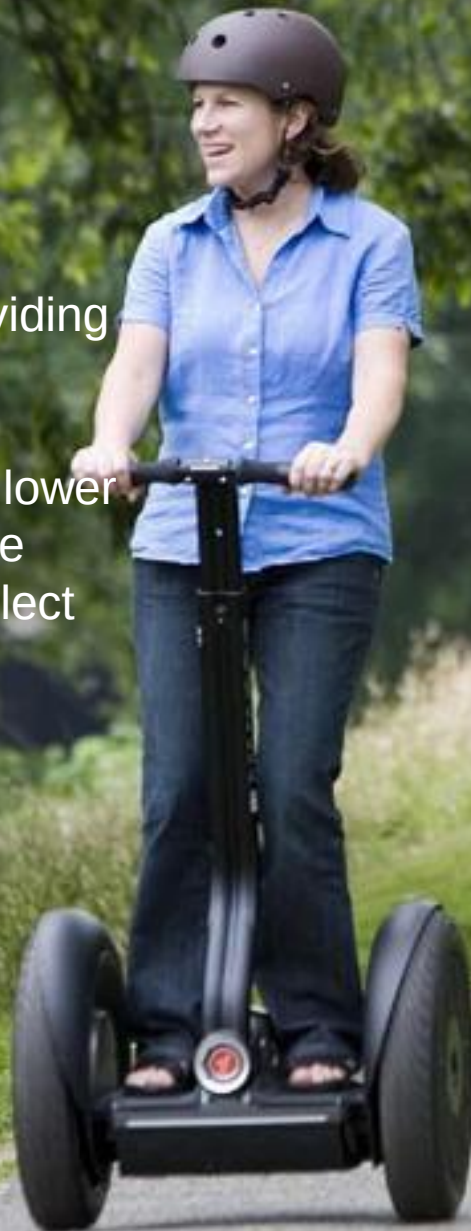
Organizations are graded on not just their operations, but on the behavior of their employees as well.

The Greenest organizations are those whose operations, value chains, products, services, and employees have the smallest aggregated impact.

Certain Green leaders, such as Whole Foods and Toyota, have mandated that employees each spend no more than 1000 Points per week (the amount which has been deemed "sustainable" by the UN 's Environmental Committee), both in and outside of work.

Employees begin to list their Eco-Points on their résumés and CVs, and these figures are validated as part of the standard background check performed by HR .

This scenario is about creating sustainable behaviour and products that support it. It is about providing solutions and rewarding the customer with lower prices the more points they collect through good solutions.



Scenario 4

Sustainable cities
and self sufficient
sustainable homes

In two decades, organic farming could restore 1/2 of soil organic matter lost due to intensive agriculture - meeting or exceeding Kyoto Protocol requirements for reducing GHG emissions.

Source: Organic Center calculations.



This scenario is about cities where communities provide their own energy, clean water, green transport, food produce and share within the community. It's about homes that grow their own, clean their own water, recycle their own waste...



Urban organic agriculture and home farms



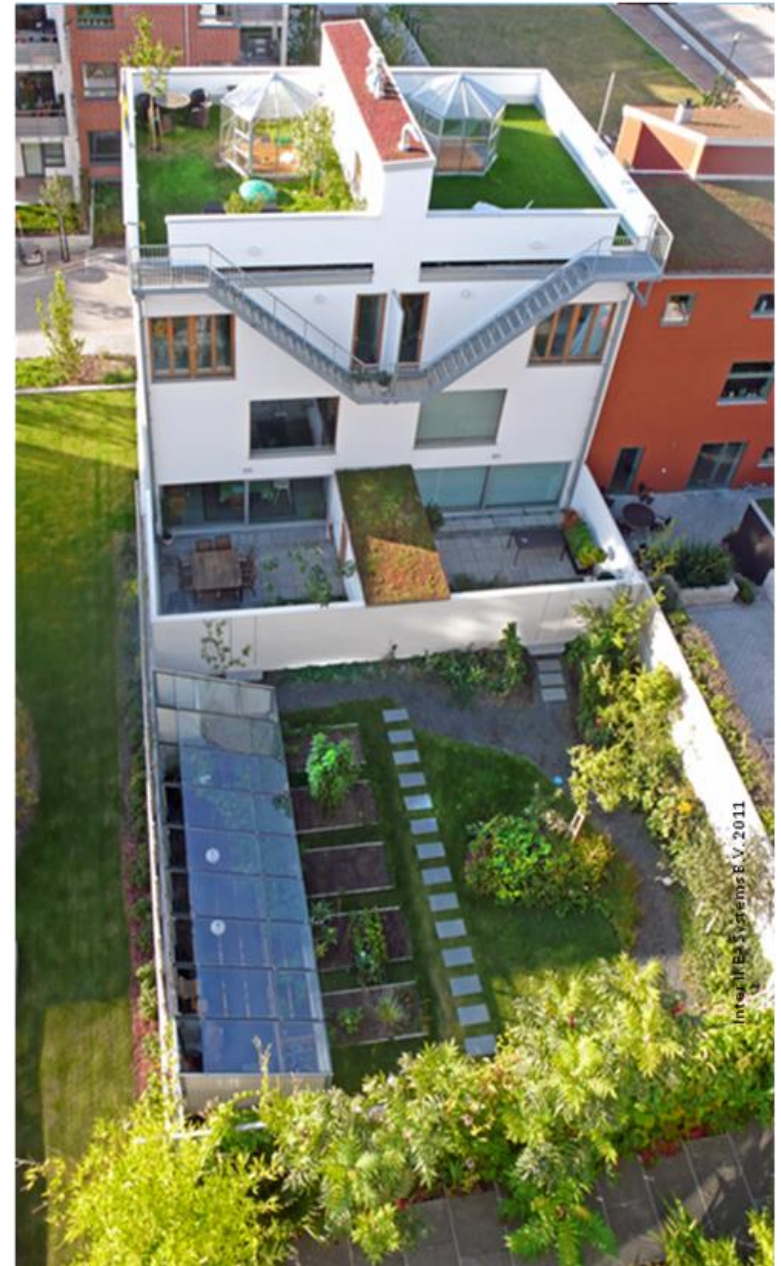
The sustainable home in the sustainable city

Working together –own expertise and knowledge with knowledgeable outside partners to look at the totality.

A home that uses less energy, water and waste, in a city that uses less energy, water and waste.

A home that is fresh, fun and healthy in cities that are fresh, fun and healthy.

Places where children can grow up and live healthy meaningful lives with a sustainable future, both for them and the planet.



The sustainable home.

Think about it.

An entire home that is sustainable!

What a wonderful challenge.

Imagine when we can say that we know that all resources used and needed in the home are within the limits for what our planet can maintain.

It is when all our products are material efficient and made from low-impact materials.

It is when we know that we have reduced and reused water consumption so efficiently that it will not impact the water cycle.

It is when we have developed energy efficient products and solutions for the home, based on renewable energy.

Imagine what this will mean to the many people when we can offer these products and solutions that are also affordable for the many.

Think about it.

(



R

Repeatable

(something you can do regularly, like every day or every week)



I

Inspirational

(something that motivates others to make a change in their lives)



S

Sustainable

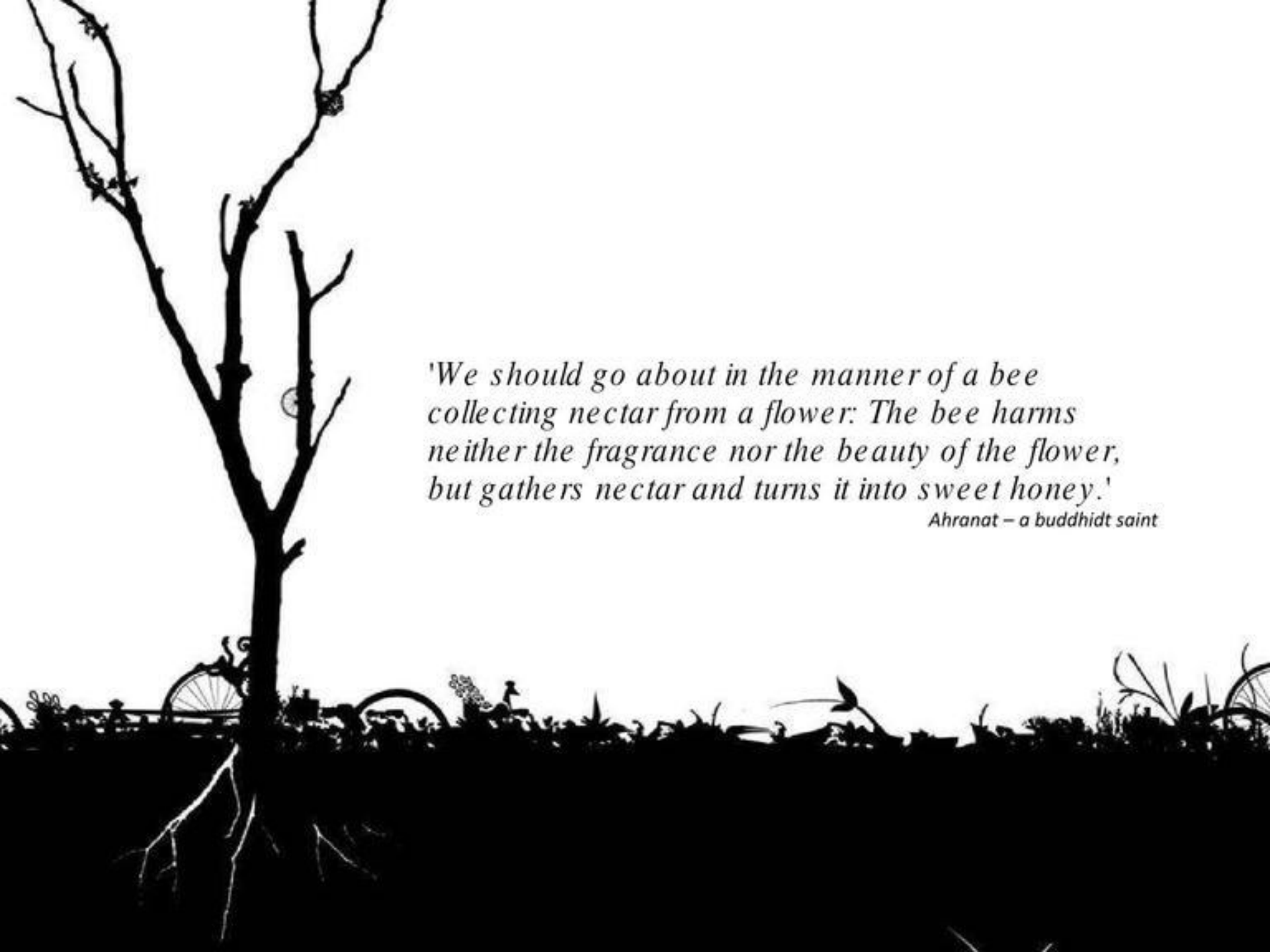
(something that keeps you and the planet healthy and well)



E

Enjoyable

(something that makes you happy)

A black and white silhouette illustration. On the left, a tall, bare tree with several branches reaches towards the top of the frame. At its base, a bicycle is parked, its wheels and frame visible. A horizontal line of various plants and flowers stretches across the bottom of the image. The background is plain white.

*'We should go about in the manner of a bee
collecting nectar from a flower: The bee harms
neither the fragrance nor the beauty of the flower,
but gathers nectar and turns it into sweet honey.'*

Ahranat – a buddhist saint